

Personalized AI assistants could either promote or hinder the emergence of collective intelligence

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Abstract

Advances in artificial intelligence (AI) have enabled the wide use of personalized assistance for cognitive tasks, which enhanced individual performance on various fields. However, the impact of AI assistance on collective intelligence remains poorly understood. AI assistance may undermine the emergence of collective intelligence by reducing the individual diversity or introducing structured biases into a population. Here, we develop a human-AI collective prediction model mediated by social learning based on incentive structure by extending the existing works on collective intelligence. Our model uncovers that emergence of collective intelligence is depend on how human and AI interact. When AI works as if it is omniscient and gives assistance regardless of users' interests, collective intelligence will be promoted. However, when AI works as a chat-bot and gives assistance only based on users' interests, collective intelligence could be hindered due to diversity loss and overconfidence on AI. We showed that, to promote collective intelligence, penalizing or incentivizing AI usage is not reliable. Instead, elaborating more diversity-aware incentives could be the key to promote and maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of personal AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

66 Main

67 Model

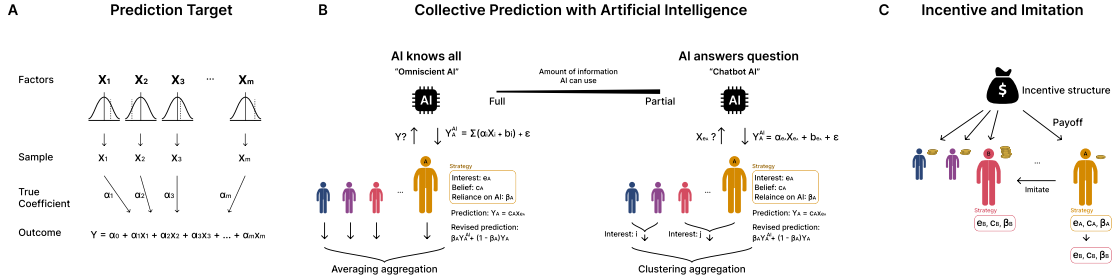


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

68 We begin by extending previously established models of CI, in which a group of
 69 players participates in a prediction task (Figure 1): each player makes an individual
 70 prediction based on partial observations of the underlying causal factors, and the
 71 group forms a collective prediction by aggregating these individual predictions [30,
 72 12, 13]. Specifically, the prediction target (Y) is modeled as a linear combination of
 73 M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor
 74 is drawn from independent normal distribution (Figure 1A). Then, each individual
 75 A observes a single factor of their interest x_{e_A} and makes a prediction based on
 76 their own belief c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then
 77 independently consult their AI assistants and revise their prediction based on their
 78 reliance on AI β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). This revised prediction
 79 is aggregated to form collective prediction. **[SWP: I think we need to say averaging**
 80 **and clustering aggregation here?]**

81 To model AI, we assume that AI has already been trained on similar tasks and is
 82 able to make predictions based on partial or full observations of causal factors (Figure
 83 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 84 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 85 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 86 amount of information AI uses when making predictions. In one extreme, the AI
 87 can observe all causal factors and use all information to make predictions, regardless

88 of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 89 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 90 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 91 and ‘Chatbot AI’, respectively.

92 Each generation consists of multiple round of predictions, and each player is re-
 93 warded a payoff according to the incentive structure (Figure 1C). For simplicity,
 94 we assume that a sufficiently large number of prediction rounds occurs within each
 95 generation, allowing us to use the expected payoff value rather than computing the
 96 payoffs for every round. We consider two incentive structures from previous work,
 97 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 98 lective accuracy in the absence of AI [13]. The feedback structure gives higher payoff
 99 to individuals who can push the collective prediction toward the true target, regard-
 100 less of their individual predictive accuracy. [SWP: *maybe we need equations...*]
 101 In contrast, the niche-expert structure gives higher payoff to individuals who have
 102 both better predictive accuracy and a minority interest. [SWP: *maybe we need*
 103 *equations... but also say that this preserves diversity*] We note collective intelligence
 104 fails to emerge under the combination of averaging aggregation with niche-expert
 105 structure in the absence of AI [13].

106 Through time, the group evolves by allowing each individual to imitate peers
 107 who have received higher payoffs (Figure 1C). Specifically, after each generation, two
 108 individuals are randomly sampled, and one imitates all three strategies (the factor of
 109 interest, factor belief, and reliance on AI) of the other with a probability determined
 110 by the difference in their expected payoffs. Individual predictions are aggregated
 111 into a collective prediction $\hat{Y}$ by simple averaging when using Omniscient AI and by
 112 clustering when using Chatbot AI. Collective accuracy is defined as the reduction in
 113 prediction error achieved by the collective prediction relative to a baseline prediction
 114 α_0 . Hereafter, we refer to the near-perfect collective accuracy as accurate CI.

115 This model now allows us to explore the impact of AI on CI depending on the
 116 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 117 of CI with and without AI and subsequently investigate a better incentive structure
 118 that can promote CI in the presence of AI.

119 AI can either promote or prevent the emergence of CI

120 To understand the impact of AI, we first compare collective dynamics with and
 121 without AI. We find that, except when Omniscient AI is used under the feedback
 122 structure, a group with AI fails to achieve accurate CI (Fig. 2). However, the effect
 123 of AI differs depending on the type of AI.

124 When Omniscient AI assists players, the group becomes equally or more accurate
 125 than the group without AI. Under the feedback structure, a group achieves accurate
 126 CI regardless of the assistance of AI (Fig. 2A). However, under the niche-expert
 127 structure, the group without AI is no better than the baseline prediction. Although
 128 Omniscient AI cannot help players achieve accurate CI, the group’s prediction with

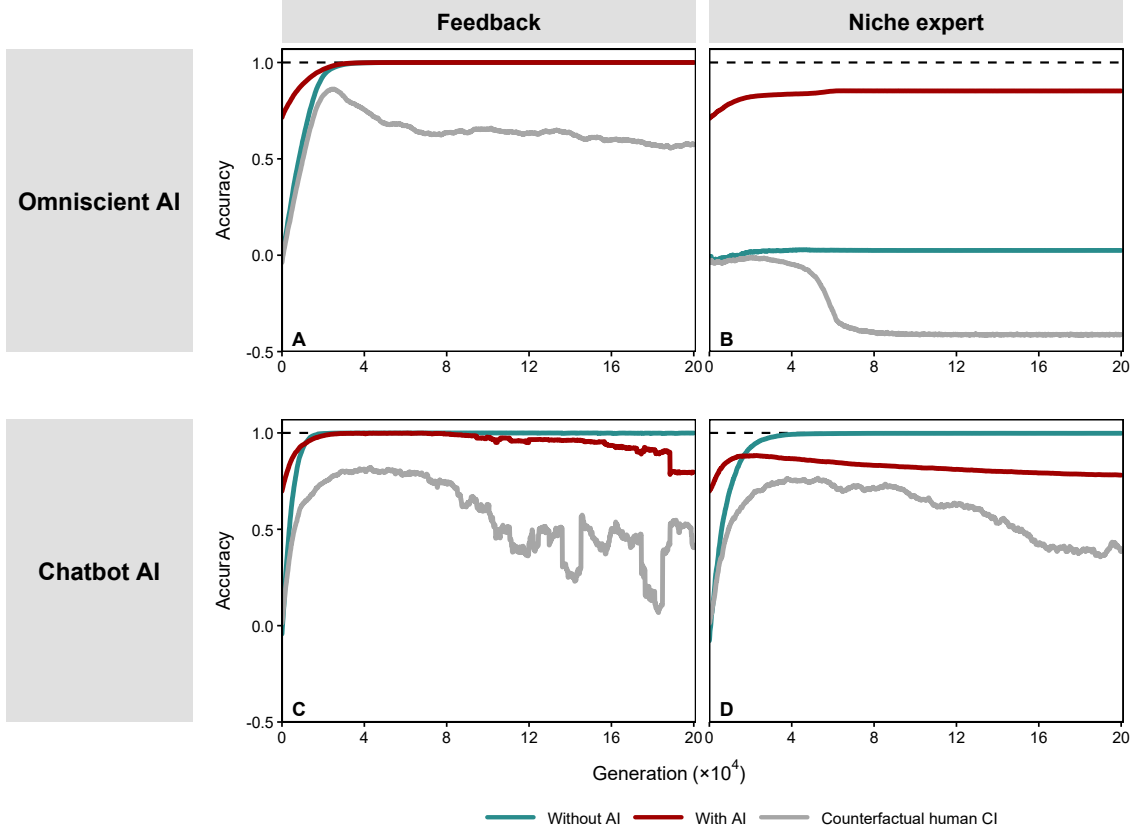


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by Omniscient AI under the feedback structure (A), by Omniscient AI under the niche-expert structure (B), by Chatbot AI under the feedback structure (C), and by Chatbot AI under the niche-expert structure (D). The green line represents the trajectory of collective accuracy without AI assistance, whereas the red line represents the trajectory with AI assistance. The gray line represents counterfactual collective accuracy, computed from players’ unrevised predictions while retaining the evolutionary trajectory of the group with AI assistance, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

AI is more accurate than both that of the group without AI and the AI itself (Fig. 2B). These results are robust across different levels of AI accuracy and can be shown mathematically by assuming a sufficiently large group of players (Fig. S2).

When Chatbot AI assists players, the group becomes less accurate than the group without AI. Although both incentive structures can achieve accurate CI without AI,

the assistance of Chatbot AI hinders group performance and prevents accurate CI from emerging (Fig. 2C, D). This occurs through different mechanisms depending on the incentive structure. Under the feedback structure, Chatbot AI leads to a loss of interest diversity, which eventually makes it impossible for the group to account for variation in some causal factors. This leads to an increase in collective variance and stochastic decrease in collective accuracy (Fig. S1). The loss of interest diversity may arise even without AI after sufficiently many generations in a finite population. However, Chatbot AI makes the group more vulnerable to the diversity loss. Under the niche-expert structure, the group becomes overly reliant on AI, causing the collective prediction to inherit the AI’s bias (Fig. S1). This leads to an increase in collective bias and a decrease in collective accuracy. The niche-expert structure may also drive reliance on AI toward zero, depending on the relationship between the true coefficient and AI bias (Fig. S2). In this case, the group completely excludes AI assistance and builds CI on its own, which substantially delays the emergence of accurate CI.

To investigate whether the group can retain its predictive ability when AI assistance suddenly becomes unavailable at a given generation, we measure counterfactual collective accuracy. This measure follows the same evolutionary trajectory as the simulation with AI, but accuracy is computed using the players’ unrevised predictions. We find that counterfactual collective accuracy is always lower than the collective accuracy of groups, regardless of the AI mode or incentive structure.

In summary, AI assistance may promote CI when it is omniscient, whereas AI operating with limited information may prevent the emergence of accurate CI. These findings underscore the need for incentive structures that can mitigate the potential risks associated with AI assistance.

Directly incentivizing or penalizing AI use can additionally harm CI

Given the strong influence of AI on collective outcomes, a straightforward intervention is to directly incentivize or penalize the use of AI. To investigate whether such direct incentives or penalties help emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of incentive ($\lambda > 0$) or penalty ($\lambda < 0$).

Directly incentivizing or penalizing AI use further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure (Fig. 3A). This is because incentivizing or penalizing AI usage pushes the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI, respectively (Fig. 3B, D, F, H). In some cases, when AI bias is relatively large and the penalty is weak, penalizing AI use can achieve higher collective accuracy than the original incentive structure. However, in most cases, penalizing AI use imposes strong selection against AI reliance, pushing

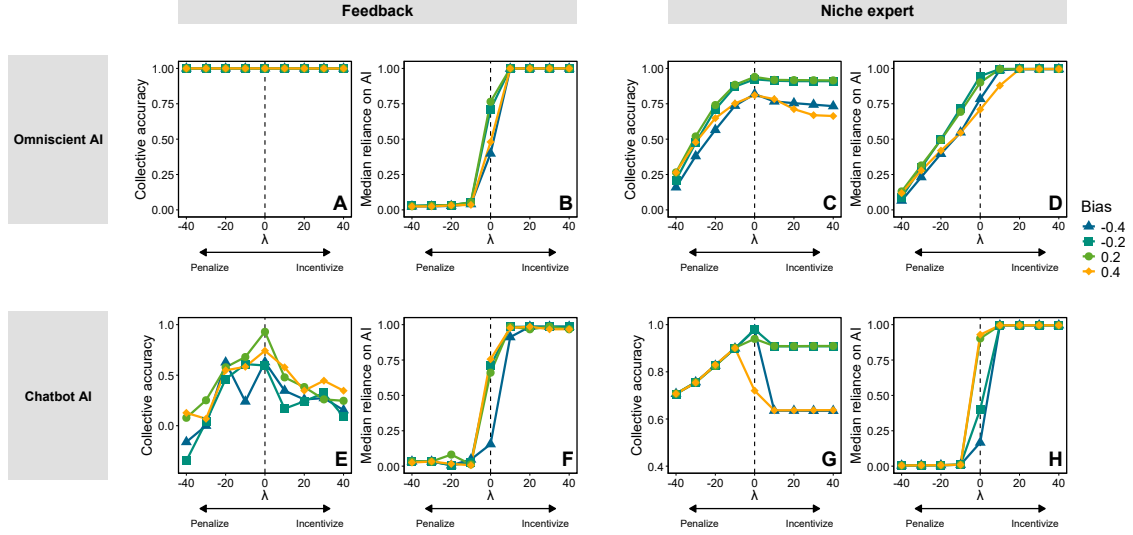


Figure 3: **Effect of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) applied to the incentive structure. Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). (A, B) Omniscient AI under the modified feedback structure, (C, D) Omniscient AI under the modified niche-expert structure, (E, F) Chatbot AI under the modified feedback structure, and (G, H) Chatbot AI under the modified niche-expert structure. Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy and median reliance on AI are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

the group toward lower stationary collective accuracy. Although the group with Chatbot AI under feedback structure shows stochastic breakdown of CI which doesn't result in stationary collective accuracy, incentivizing or penalizing AI use further strengthens the breakdown (Fig. 3E). This result suggests that simply incentivizing or penalizing the AI use may exacerbate rather than resolving the problem of AI.

Balanced incentives mitigate AI-induced failures of CI

Neither the original incentive structures nor simple interventions targeting AI use are sufficient to promote CI. What incentive structure, then, would work in the presence of AI? Focusing on the Chatbot AI, in which proposed incentive structures fail to promote accurate CI, we investigate whether a better structure can overcome the risks introduced by AI. Here, we propose a “balanced” structure, defined as the simple average of the feedback and niche expert incentives.

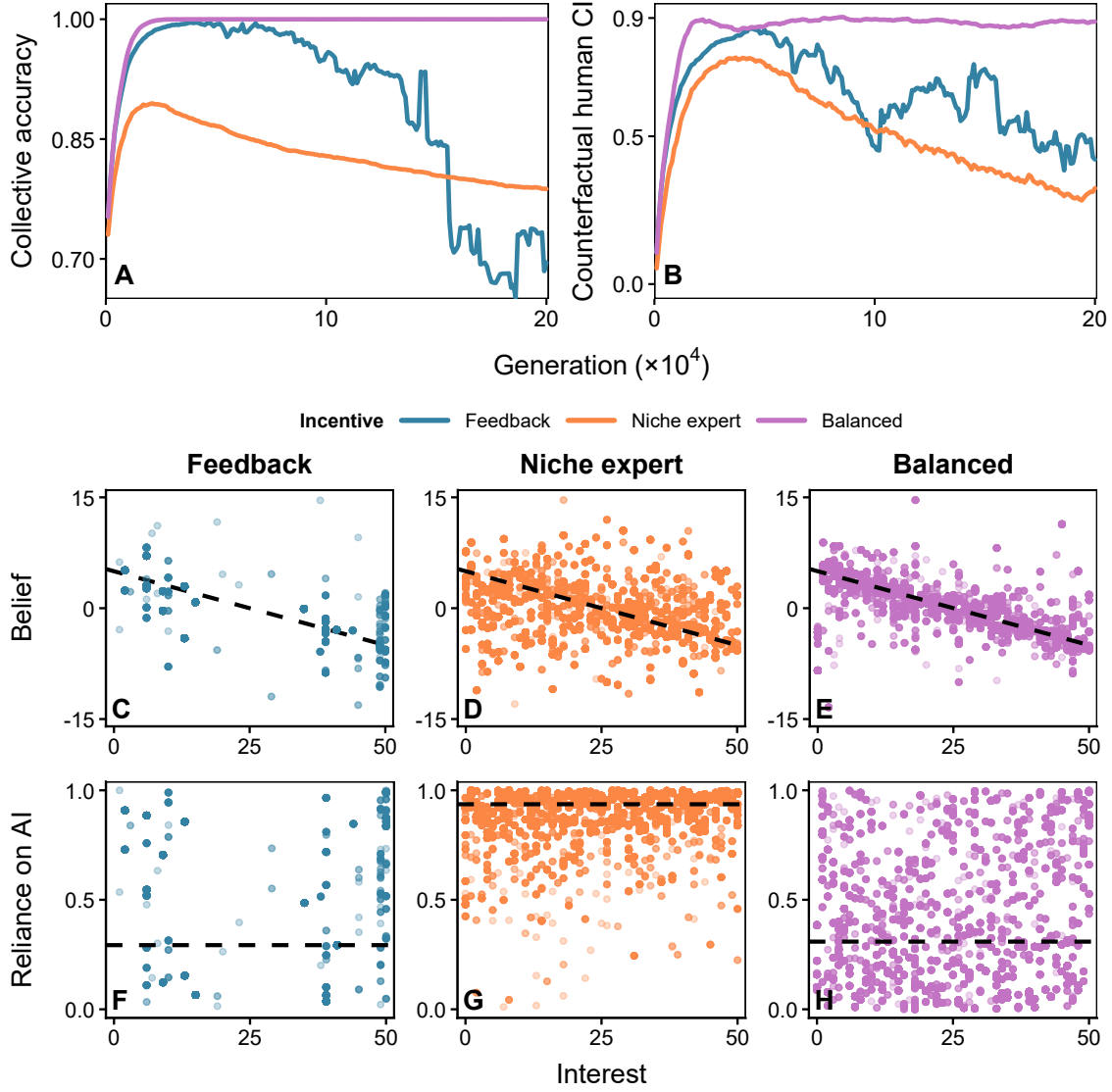


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents a player. The dashed lines in (C-E) represent the true coefficients (α_i) of the prediction target, and the dashed horizontal lines in (F-H) indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

185 Core idea behind the balanced structure is to maintain interest diversity while
 186 preventing the collective bias of the group. The feedback structure is prone to in-

terest diversity loss, whereas can strongly prevent collective bias by incentivizing players whose predictions lie opposite to the collective prediction. By contrast, the niche-expert structure can maintain interest diversity because it directly incentivizes minority interests. However it cannot prevent collective bias because, within the same factor of interest, individuals are rewarded only according to their individual predictive performance, which homogenize the predictions.

By integrating both structure, the balanced structure can promote accurate CI by maintaining both interest diversity and preventing collective bias (Fig. 4, S3). Compare to the feedback and niche-expert, players are both evenly distributed through factors, and beliefs are close to the real coefficient (Fig. 4). Moreover, counterfactual collective accuracy maintained high compare to both feedback and niche-expert. These patterns are driven by the emergent property of the balanced structure, in which group increases its reliance on AI as the AI becomes more accurate, demonstrating appropriately calibrated AI use.

Discussion

In this paper, we study how AI and incentive structures affect the emergence of CI. Our model suggests that the effect of AI depends on the type of AI used by the group. AI that can incorporate all relevant causal factors may improve group performance, whereas AI that relies on users to gather information about the causal factors may impair it. By examining different variants of incentive structures, we further show that direct interventions targeting AI use generally harm group performance. Instead, we suggest balanced structure which can promote CI while also generating appropriately calibrated AI use as an emergent property.

Although Omniscient AI can produce group performance that is equal to or better than that achieved without AI, such AI may be difficult to realize in practice because it needs to be able to observe causal factors independently. Achieving this capability would require hardwares, such as sensors, as well as autonomous data acquisition capabilities. For this reason, most AI services currently available are better represented by the Chatbot AI in our model, which highlights the potential risks of AI use for CI.

In our model, individuals revise their predictions based on the AI’s assistance. It is therefore unsurprising that AI assistance reduces the diversity of individual predictions, a pattern that has also been observed empirically. However, a reduction in prediction diversity does not necessarily lead to the collapse of CI. The principal risk arises when the incentive structure fails to guide the group toward an appropriate set of strategies. The feedback structure leads to the breakdown of CI because it recruits more players to more important factors, which have larger coefficients. This characteristic creates a risk of interest-diversity loss in finite populations, a situation in which no player remains interested in certain factors because most players shift toward relatively important factors. Without AI, this occurs when initial in-

dividual predictions are not sufficiently diverse or when there is no source of novel ideas, such as mutation. Revising individual predictions with AI assistance exacerbates this problem by pulling individuals’ predictions toward the AI’s prediction, thereby reducing prediction diversity even under the same belief distribution. The niche-expert structure hinders CI because it favors extreme levels of reliance on AI. Because individuals within each interest group are affected only by their own performance, incentives based on individual performance reach their maximum at either full reliance on AI or no reliance at all. This extreme reliance results in suboptimal collective accuracy or substantially delays the emergence of CI.

Because AI already generates substantial benefits as well as costs, organizations increasingly seek to improve collective performance by regulating AI use. Such interventions may take direct forms, including AI-use policies or token-based leaderboards, or may operate indirectly through other incentives, such as productivity rewards or the social stigma associated with AI use. However, our model suggests that directly incentivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups to propagate AI biases, whereas penalizing AI use could lead to underreliance, preventing groups from using AI even when it performs well in the relevant domain. Our result instead shows that appropriate AI use and accurate CI can be promoted by focusing on the classic properties of groups that have been emphasized for decades—maintaining diversity and preventing collective bias—rather than on AI use itself.

The consistently limited counterfactual collective accuracy observed across AI types and incentive structures suggests that the presence of AI can degrade human group’s own performance, consistent with empirical evidence. This may reduce collective performance when AI assistance becomes unavailable because of resource constraints, system failure, or model collapse, forcing the group to rebuild CI through further adaptation. The higher counterfactual collective accuracy achieved under the balanced structure indicates that an appropriate incentive structure can mitigate the risk of CI breakdown following the loss of AI assistance.

Our model and analysis have several limitations. First, the model considers only social learning among humans and treats the influence of AI as unidirectional. In reality, humans can learn from AI, while AI systems can also be further trained on human-generated data. Such bidirectional adaptation may further reduce group performance by increasing the similarity between human and AI predictions and thereby weakening their complementarity. Empirical studies of human–AI collaboration likewise suggest that stronger correlations between human and AI predictions can diminish the benefits of collaboration. Second, we model AI errors using factor-specific additive biases and random error. Previous studies of AI bias and error suggest that additive biases may arise from skewed training data, or systematic calibration. However, AI errors may also exhibit more complex structures, including biased estimates of factor effects and correlations across prediction errors. Further empirical research is therefore needed to characterize the structure of AI bias and

error, together with modeling approaches that can incorporate these more complex patterns. Finally, to retain the tractability of established CI models, we assume that the prediction target is a linear combination of causal factors, which does not hold for many real-world tasks. In particular, many machine learning methods used in AI systems are capable of capturing nonlinear relationships. Future research should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

where the fixed coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 is a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel prediction task. In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal factor X_{e_A} and predicts the target by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player's prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player's strategy additionally includes reliance on AI, denoted by β_A . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. Although the bias is indexed by factor, all b_i , including b_0 , are assigned the same value in the main analyses for simplicity. The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore held τ constant and varied the magnitude of the additive bias to manipulate the level of AI accuracy across analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research, depending on the type of AI used by the group.

When the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

335 When the group is assisted by the Chatbot AI, both the player’s initial prediction
 336 and the AI’s prediction are based on the same causal factor. Each revised predic-
 337 tion therefore contains information about only one factor and can be treated as an
 338 estimate of that factor’s contribution to the target. We consequently use clustering
 339 aggregation.

340 Clustering aggregation is performed by grouping individuals according to their
 341 interests and calculating the mean prediction within each group. The collective
 342 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

343 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number
 344 of players in that set.

345 When the group is not assisted by AI, collective predictions are computed from the
 346 unrevised individual predictions throughout the simulation. Both simple averaging
 347 and clustering can be used in the absence of AI. For fair comparison, we use the
 348 same aggregation rule as in the corresponding AI-assisted condition.

349 Incentive structures

350 Players receive payoffs determined by an incentive structure based on their strategies.
 351 We consider five incentive structures: feedback structure, niche-expert structure,
 352 direct intervention on AI use of these two structures, and a balanced structure.

353 The feedback structure does not directly reward individuals according to their
 354 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 355 residual error of the collective prediction and therefore have the potential to move
 356 the collective prediction toward the true value. Given player A ’s revised prediction
 357 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

358 The niche-expert structure assigns higher payoffs to individuals who make accu-
 359 rate predictions while having an underrepresented interest. Specifically, the squared
 360 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 361 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

362 Previous research has shown that these two incentive structures can produce
 363 nearly perfect CI through social learning, except when the niche-expert incentive is
 364 combined with simple averaging. They are therefore used as baseline structures for
 365 examining how AI assistance affects the emergence of CI.

366 To examine the effects of directly encouraging or discouraging AI use, we mod-
 367 ify the feedback and niche-expert structures by adding a term proportional to the
 368 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

369 Here, λ is a weighting coefficient representing the strength of the direct incentive
 370 or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a
 371 negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied,
 372 and the payoff functions reduce to the original feedback and niche-expert structures.

373 The balanced structure is defined as the arithmetic mean of the feedback and
 374 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

375 Simulations

376 All simulations were conducted using an agent-based model with 50 causal factors
 377 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 378 target were sampled at the beginning of the simulations and subsequently held fixed.

379 The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in
 380 the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization.
 381 The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

382 Players’ initial strategies were also sampled from predefined probability distri-
 383 butions. The distribution of individual belief c_A depended on the aggregation rule.
 384 Under simple averaging, each factor’s contribution is scaled by the proportion of
 385 players interested in that factor. We therefore sampled beliefs from a Gaussian dis-
 386 tribution with mean zero and standard deviation 100, following the scaling used
 387 in previous work [13]. Under clustering aggregation, beliefs were sampled from a
 388 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 389 are averaged separately within each interest group and then summed, no additional
 390 scaling of c_A is required. The initial interest of each player was sampled uniformly
 391 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
 392 distribution on $[0, 1]$.

393 We assumed that prediction events occurred on a substantially faster timescale
 394 than social learning. At each generation, we therefore calculated the expected payoff
 395 $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events.
 396 The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$
 397 and the stochastic AI error ϵ , conditional on the current population strategies and
 398 the fixed target parameters. Closed form expression of expected payoffs were derived
 399 for all incentive structures and are presented in the Supplementary Methods.

Each generation consisted of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , were randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduced a mutation rate μ , such that with probability μ , player A did not imitate player B but instead redrew all strategy components from the same distributions used to initialize the population. The mutation rate was set to zero in all simulations except those presented in Supplementary Figure 4.

Unless otherwise specified, each simulation was run for 200,000 generations. When a result was represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we defined six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances were evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

Collective accuracy was normalized relative to the baseline prediction α_0 . A negative value indicates that the collective prediction performs worse than the baseline. Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining the evolved interests and beliefs of the AI-assisted population. In calculating interest diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Data and code availability

All data are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/collective-intelligence-with-AI>).

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Contributions

Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing - Original Draft Preparation, Writing - Review &

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

433 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
434 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

435 **Competing Interests**

436 The authors declare no competing interests.

Supplementary Text

1 Omniscient AI

Omniscient AI assumes that AI can incorporate all causal factors regardless of the player's interest.

1.1 Model

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (S1)$$

For simplicity, prediction of Omniscient AI can be written as:

$$Y^{AI} = Y + B + \epsilon \quad (S2)$$

where Y stands for true prediction target ($Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M$) and B is sum of all fixed biases ($B = b_0 + b_1 + \cdots + b_M$), ϵ follows normal distribution with zero mean, τ standard deviation with no correlation within or between players. Player A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (S3)$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

where c_A stands for player A's belief on the factor, e_A stands for the interest, and β_A stands for reliance on AI.

1.2 Aggregation

Because Y_A^{AI} is subject to all M factors, we use simple averaging aggregation for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \frac{1}{N} \sum_{j=1}^N Y_j^{re} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j Y^{AI} + (1 - \beta_j) Y_j \} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j} \} \end{aligned}$$

Reparameterization for simple derivation and interpretability.

$$\bar{\beta} := \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S5)$$

454

$$\bar{\eta} := \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S6})$$

455

$$\gamma_i := \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S7})$$

456 $\bar{\beta}$ can be interpreted as mean reliance on AI, γ stands for the coefficient for the factor
 457 i which is solely based on players' belief.

458 By this reparameterization, we can write $\hat{Y}^{re}$ as follows.

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S8})$$

459 1.3 Expected payoffs

460 Feedback

$$\delta_0 := \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0) \quad (\text{S9})$$

461

$$\delta_i := \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0) \quad (\text{S10})$$

462

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S11})$$

463 See equation Eq. (S6) for the $\bar{\eta}$.

464 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \\ &= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \end{aligned}$$

465 **Niche-expert**

466 If $e_A = 0$, player does not observe causal factor and their unrevised prediction is same
 467 with their belief c_A . Such that, the expected payoff is

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \\
 &= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

468 Else,

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \\
 &= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B + \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

469 **Incentivizing or penalizing AI use**

470 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
 471 payoff as

$$\mathbb{E}(\pi_{A, Feedback}) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] + \lambda\beta_A. \quad (S12)$$

$$\mathbb{E}(\pi_{A, Niche-expert}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (S13)$$

472 **1.4 Expected accuracy**

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (S14)$$

473

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S15})$$

474

$$\begin{aligned} R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\ &= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \end{aligned}$$

475 2 Chatbot AI

476 2.1 Model

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon \quad (\text{S16})$$

477 b_{e_A} stands for factor specific additive bias, ϵ follows normal distribution with zero
 478 mean, τ standard deviation with no correlation within or between players. Player
 479 A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S17})$$

480

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S18})$$

481 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
 482 β_A stands for reliance on AI.

483 2.2 Aggregation

484 For Chatbot AI, clustering aggregation is used for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A Y_A^{AI} + (1 - \beta_A) Y_A\} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon] \end{aligned}$$

485 Reparameterize for the simplicity.

$$\tilde{c}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \quad (\text{S19})$$

486

$$\tilde{B}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A} \quad (\text{S20})$$

487

$$\bar{B} := \sum_{i=0}^M \tilde{B}_i \quad (\text{S21})$$

488

$$\tilde{\eta}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon \quad (\text{S22})$$

489

$$\bar{\eta} := \sum_{i=0}^M \tilde{\eta}_i \quad (\text{S23})$$

490 2.3 Expected payoffs

491 Feedback

492

$$\delta_0 := \alpha_0 - \tilde{c}_0 - \bar{B} \quad (\text{S24})$$

493

$$\delta_i := \alpha_i - \tilde{c}_i, \quad (i \neq 0) \quad (\text{S25})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S26})$$

494 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \end{aligned}$$

495 Niche-expert

496 If $e_A = 0$,

$$\begin{aligned} \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\ &= -\mathbb{E} \left[\rho_0 \left[\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \\ &= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \\ &\quad \left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \end{aligned}$$

497 Else,

$$\begin{aligned}
\mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \\
&= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1 - \beta_A)c_A\}X_{e_A} + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i\right]^2\right] \\
&= -\rho_{e_A}\left\{(1 - \beta_A)^2(c_A - \alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2\right. \\
&\quad \left.+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2 \tau^2\right\}
\end{aligned}$$

498 Incentivizing or penalizing AI use

499 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
500 payoff as

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A. \quad (\text{S27})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S28})$$

501 Balanced

502 Expected payoff of the balanced structure can also be driven by arithmetic mean of
503 expected payoff of feedback and niche-expert.

$$\mathbb{E}(\pi_A) = \mathbb{E}\left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}\right] = \frac{\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - \mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right]}{2} \quad (\text{S29})$$

504 2.4 Expected accuracy

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S30})$$

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S31})$$

$$\begin{aligned}
R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\
&= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}
\end{aligned}$$

507 3 Evolutionary dynamics

508 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
509 evolutionary dynamics using the replicator equation. Let λ denote a strategy, p_λ its
510 frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. The replicator equation is

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S32})$$

511 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
512 with positive frequency have the same expected payoff. A boundary state can also
513 be stationary because a strategy with exactly zero frequency cannot increase under
514 the replicator equation. Thus, stability of a boundary state needs to be considered
515 separately by examining invasion by a rare strategy.

516 3.1 Omniscient AI

517 3.1.1 Feedback

518 When the population size approaches infinity, the effect of stochastic error on the
519 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
520 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research [13], we define the
521 Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S33})$$

522 The function V is non-negative and equals zero when the collective prediction is
523 identical to the target. To examine how V changes under the evolutionary dynamics,
524 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

525 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S34})$$

526 Although K changes with the population state, it is common to all strategies at a
527 given state and can therefore be treated as a constant with respect to the strategy
528 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S35})$$

529 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S36})$$

530 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S37})$$

531 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K\beta + (1 - \beta) c G_e, u).
\end{aligned}$$

532 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
533 λ is

$$u_{\lambda} = K\beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S38})$$

534 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S39})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S40})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S41})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S42})$$

568 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 569 the expected squared prediction error at the corresponding boundary.

570 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S43})$$

571 Substituting this into the expected payoff gives a strictly concave function of β ,
 572 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S44})$$

573 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S45})$$

574 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S46})$$

575 We denote the optimal strategy, or the payoff-equivalent strategy class when
 576 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 577 be stationary if the optimal strategy is absent, but it is unstable against invasion
 578 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 579 a stable state can contain only S_e within each represented interest.

580 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 581 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S47})$$

582 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 583 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \dots = \rho_M^* L_M. \quad (\text{S48})$$

584 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S49})$$

585 This internal stationary state is locally stable. At equilibrium, all represented
 586 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 587 interest increases slightly above its equilibrium value, its payoff decreases because
 588 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
 589 an interest decreases below its equilibrium value, its payoff increases. The replicator
 590 dynamics therefore generate a restoring force that reduces small deviations in interest
 591 frequencies and drives the population back toward the internal stationary state.

592 If one or more interests are absent, the resulting boundary state can be stationary
 593 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 594 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 595 population is negative when $L_e > 0$, so the rare interest increases when introduced
 596 at sufficiently small frequency.

597 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 598 prediction strategies and their frequencies. Whether the constrained optimum has
 599 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 600 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 601 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 602 interest frequency remain uniquely determined.

603 This equilibrium also allows a direct comparison with the expected accuracy of
 604 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S50})$$

605 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 606 fore, because L_e is the minimum individual expected squared prediction error within
 607 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S51})$$

608 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S52})$$

609 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S53})$$

610 The collective prediction is the average of individual revised predictions. By
 611 convexity of the squared error,

$$\mathbb{E} [(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S54})$$

612 Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{\text{AI}})^2]. \quad (\text{S55})$$

613 Because the collective accuracy and AI accuracy use the same normalization by
 614 $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is
 615 strictly greater than the expected accuracy of the AI prediction.

616 Although the niche-expert structure improves collective prediction relative to the
 617 AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero
 618 factor i , the residual coefficient of the stationary collective prediction can be written
 619 as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S56})$$

620 At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore,
 621 for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S57})$$

622 Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error.
 623 Hence,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] > 0, \quad (\text{S58})$$

624 and consequently the stationary collective accuracy satisfies $R^* < 1$.

625 3.2 Chatbot AI

626 3.2.1 Feedback

627 With Chatbot AI, the effect of stochastic error on the collective prediction depends
 628 on n_i , requiring the additional assumption that $n_i \rightarrow \infty$ for every interest for the
 629 stochastic error to vanish. However, Under this assumption, the loss of interest
 630 diversity cannot occur, and therefore the stochastic breakdown of collective accuracy
 631 observed in finite populations cannot be analytically captured. Instead, by assuming
 632 that each interest is represented by infinitely many players, we can show that the
 633 expected collective prediction error monotonically decreases until perfect accuracy
 634 is achieved. Because finite population dynamics are more relevant to the empirical
 635 populations considered in this study, we retain finite population simulations for the
 636 main result.

637 3.2.2 Niche-expert

638 We first consider strategies having the same interest by optimizing the expected
 639 payoff with respect to c and β . For a nonzero interest e , the expected payoff is
 640 strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S59})$$

641 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 642 unconstrained stationary point is

$$\tilde{\beta}_e = \frac{2\alpha_0 b_e}{2b_e^2 + 2\tau^2}. \quad (\text{S60})$$

643 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 644 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Because we
 645 assumes AI model with moderate accuracy, b_e and τ is relatively smaller than α_0
 646 which makes $\tilde{\beta}_e > 1$ when $\alpha_0 b_e > 0$ and $\tilde{\beta}_e < 0$ when $\alpha_0 b_e < 0$. These boundary
 647 cases lead to extreme reliance on AI or complete exclusion of AI which both hinders
 648 the emergence of CI. Only when the prediction task is centered near zero and AI is
 649 relatively inaccurate ($\alpha_0 b_e < b_e^2 + \tau^2$), and $\alpha_0 b_e > 0$, the internal optimum β_e^* can be
 650 exist.

651 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} b_0. \quad (\text{S61})$$

652 Substituting this into the expected payoff gives a strictly concave function of β , with
 653 the unique optimum

$$\beta_0^* = 0. \quad (\text{S62})$$

Supplementary Figure

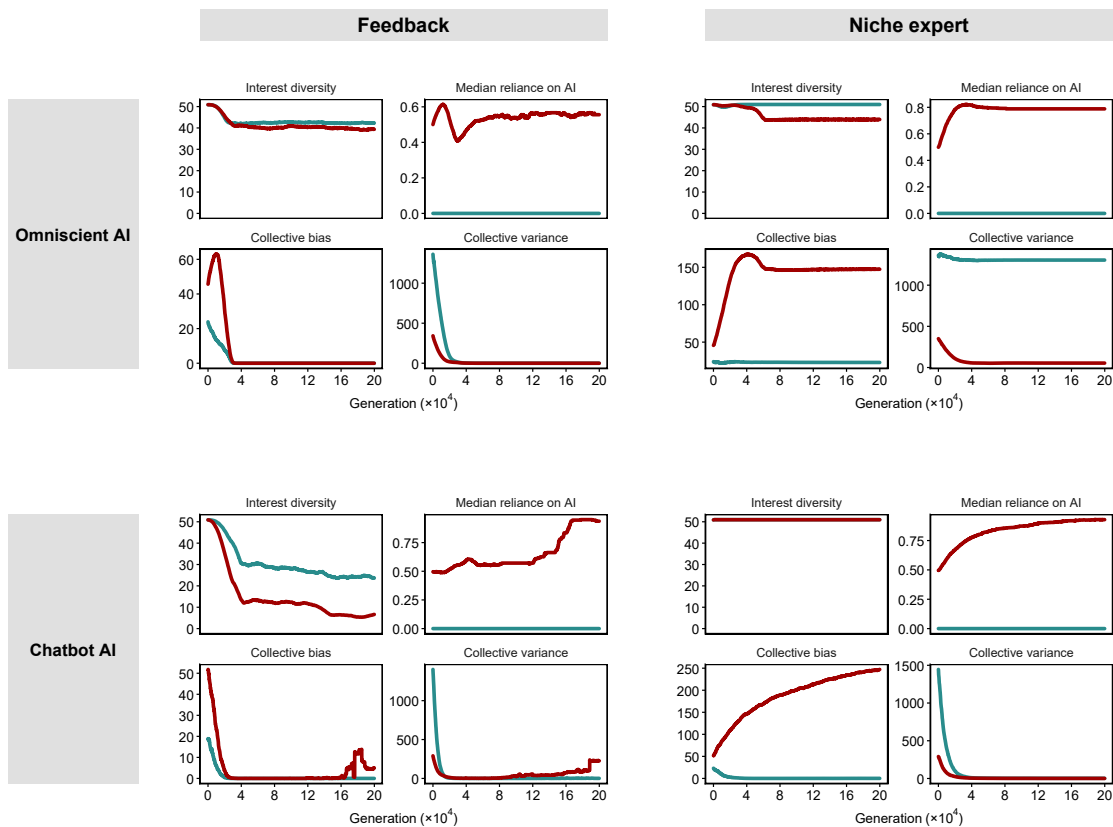


Figure S1: Caption. Text.

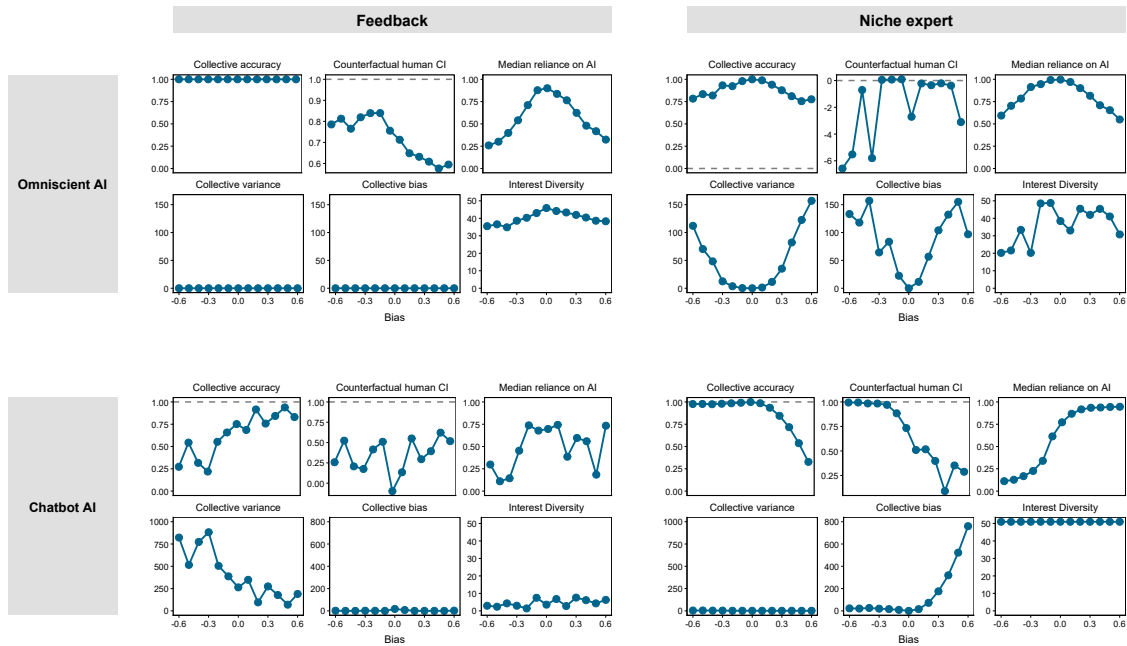


Figure S2: Caption. Text.

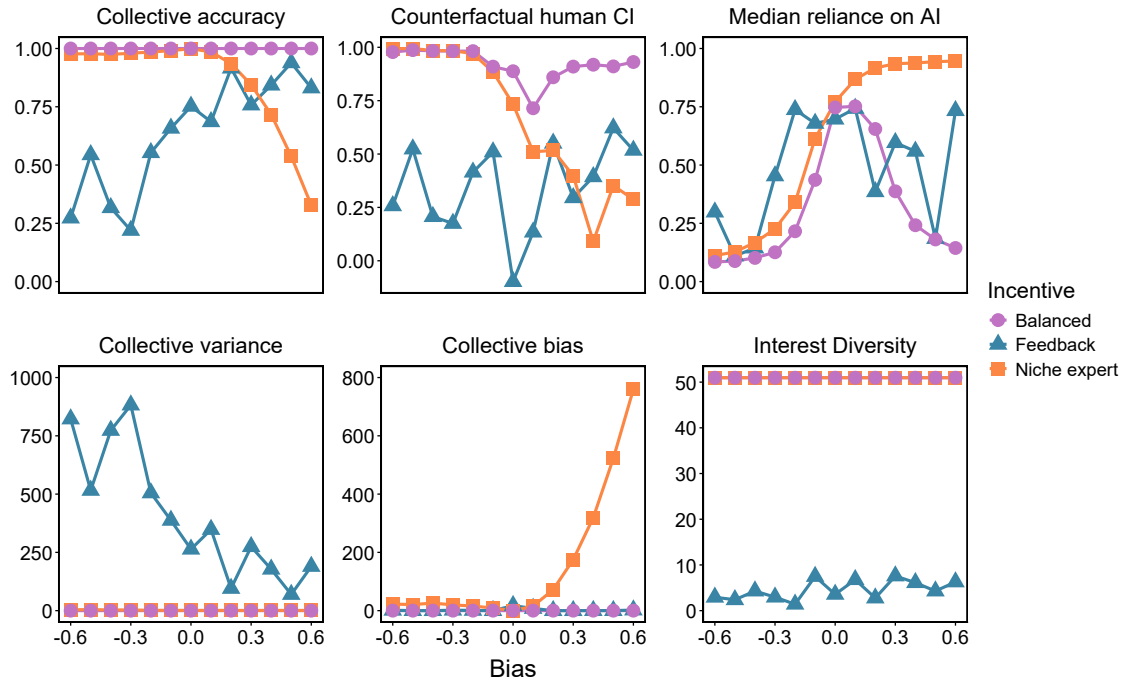


Figure S3: **Caption.** Text.

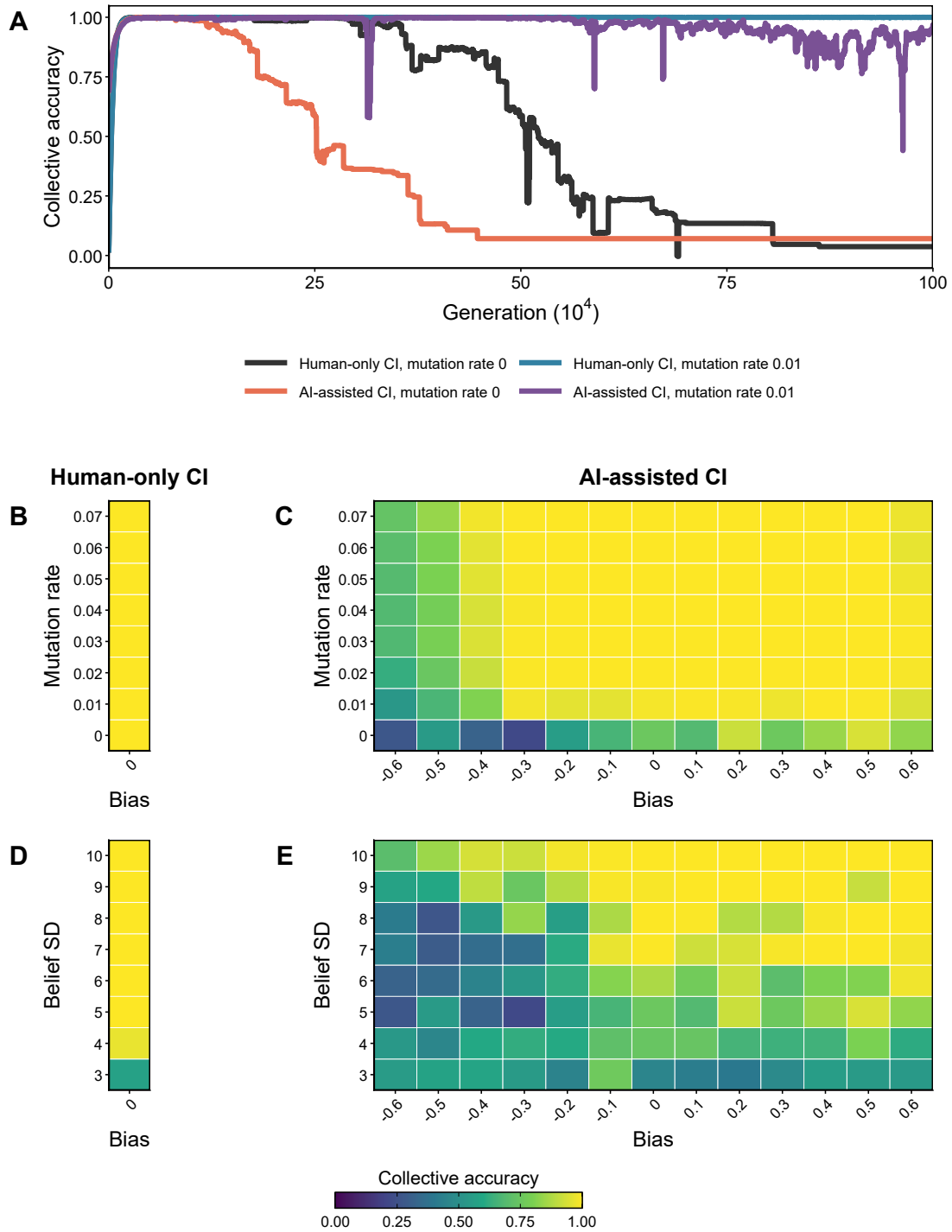


Figure S4: **Caption.** Text.

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